# AT10D-MK User Manual V1.0

Updated by 2025-07-12

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# Preface

**Thank you for choosing the company's AT10D-MK E-Lock, please carefully read the instructions before operating.**

## ◆ Disclaimer：

- Read this user manual carefully please. When you start use this product, then you are deemed to have read this user's manual.
- This Product used as assistant tool for Security only, can't prevent all kinds of deliberately theft or malicious damage vehicles. For the safety of your assets, you still need keep necessary vigilance and security awareness after you installed this product. We do not bear responsibility to any loss except product itself. Thanks!

# 1.Product Introduction

AT10D-MK is a configurable GPS solar master Elock, combined with larger rechargeable battery, support DC & solar charging, communicate base on 4G network and LoRa, with IP67 water proof that enables reliable tracking and monitoring of any fixed or mobile asset, widely used for container transportation.
AT10D-MK support LoRa can sync and work with AT10S/AT10SK slave lock.

**Product Features:**

- 2G/4G optional.
- Built-in larger Rechargeable battery.
- Support Lock ON/UNLOCK alarm.
- Support Lock rope cut alarm.
- Support casing tamper alarm.
- Support battery low power alarm.
- Support Solar charging.
- Support different sleep mode.
- Support IP67 water proof standard.
- Support Bluetooth APP configuration.
- Built-in LoRa, support working with slave Elock.

**Note:**

- ✓ Need sunshine to get solar charging;
- ✓ Please use the 5V/2A charger from the factory.

# 2.Packing List

| No. | Name | Count |
|---|---|---|
| 1 | AT10D-MK Lock device | 1 |
| 2 | Lock string wire | 1 |
| 3 | RFID card | Optional |
| 4 | Configure cable | Optional |
| 5 | DC charger | Optional |

# 3.Technical Parameters

| Name | Parameters |
|---|---|
| 2G | M25<br>2G: 850/900/1800/1900 MHz |
| 4G(Optional) | EC200N-CN<br>LTE-FDD: B1/B3/B5/B8/B28/B66<br>LTE-TDD: B34/B38/B39/B40/B41<br>GSM:B3/B8<br><br>EC21-AUX<br>LTE-FDD: B1/B2/B3/B5/B7/B8/B28<br>LTE-TDD: B40<br>WCDMA: B1/B2/B4/B5/B8<br>GSM: B2/B3/B5/B8<br><br>EC200U-EU<br>LTE-FDD: B1/B3/B5/B7/B8/B20/B28<br>LTE-TDD: B38/B40/B41<br>GSM: B2/B3/B5/B8 |
| GPS chip | (Zhongke Microelectronics) AT6558R<br>Acquisition sensitivity: -148dBm<br>Tracking sensitivity: -162dBm<br>GPS frequency: L1, 1575.42MHz<br>Locate Sensitivity: <5m,CEP<br>Hot start(average): <1s<br>Cold start(average): <30s |
| Sleep mode Current | Around mA@3.7V battery |
| Back-up Battery | 15000mAh rechargeable lithium battery |
| Size | 248mm*119mm*65mm |

| | |
|---|---|
| Tracker weight | 1336g |
| Working temperature (High Temp battery) | -25℃ - +75℃ |
| Moisture | 5% - 95% |
| LED Indicator | Use red/green/blue/green/(red/orange/green) seven color LED indicator to show the status of GSM/GPS/Iridium/Lora/power(charging/low power/charge full)status. |
| RFID Frequency | 125kHz |
| Solar panel | 6V 1.3W Solar panel |
| Solar panel Typical Electrical Characteristics | Solar Cells: Mono crystalline Solar Cell;<br>STC Conditions: 1000W/㎡,<br>1.5AM and Cell Temperature 25℃;<br>Efficiency: 21%;<br>Max Power Output: 1.8W±5%;<br>Vmp: 5.5V±5%;<br>Imp: 236mA±5%;<br>Voc: 7.2V±5%;<br>Isc: 250mA±5%;<br>Operating Temperature: -40℃ to +80℃;<br>Storage Temperature: -20℃ to +25℃(Humidity not greater than 60%);<br>Power Tolerance: -5% to 5%;<br>Cell Quantities: 22 pieces;<br>Battery conversion efficiency (%): 22%. |
| Solar panel Physical Characteristics | Module Dimension: 115.5x81.5x4.5mm;<br>Tolerance of Dimension: ±0.2mm;<br>Tolerance of Thickness: ±0.2mm. |
| **LoRa(M-HL10)** | |
| Frequency | 410~525MHz |
| RF Chip | SX1268 |
| Receiving sensitivity | -141dBm,@470MHz,125K bandwidth,SF=12 |
| bandwidth | 125K |
| Distance | 3Km,@open area,125K bandwidth,SF=12 |
| Emission current | 120mA@Maximum transmit power 22dBm |
| Receiving current | 10mA @Continuous reception mode |
| Listening current | 8mA,@ CAD detection mode |
| Sleep current | 2uA, support serial port wake-up |

| | |
|---|---|
| Bluetooth | ✓ 32-bit ARM Cortex-M4 core + FPU<br>✓ 2.4 GHz RF transceiver, support BLE5.0<br>✓ High receiver sensitivity (–94dBm@BLE)<br>✓ Programmable transmitter power, up to +3dBm<br>✓ Built-in Balun/matching network<br>✓ Receive power consumption: 3.5mA@3.0V (DCDC)<br>✓ Transmit power consumption: 3.6mA@3.0V/0 dBm (DCDC) |

# 4. Define of LED indicator

| No. | LED | Color | Status | |
|---|---|---|---|---|
| 1 | 4G | Red | Blinking | Finding SIM/Searching the 4G/GSM signal |
| | | | Light on static | Get GPRS signal/Online |
| | | | OFF | 4G module in sleep/closed |
| 2 | GPS | Green | Blinking | Finding GPS signal |
| | | | Light on static | Get GPS signal |
| | | | OFF | GPS in sleep |
| 3 | 2G | Red | Blinking | Finding SIM/Searching the GSM signal |
| | | | Light on static | Get GSM signal/Online |
| | | | OFF | 2G module in sleep/closed |
| 4 | Unlock | Blue | ON | Unlock |
| | | | OFF | Lock on/power off |
| | Low power | Yellow | Blinking | Low power |
| 5 | Charging | Orange | ON | Charging |

| | | | | |
|---|---|---|---|---|
| | Charge full | Green | On | Fully Charged |

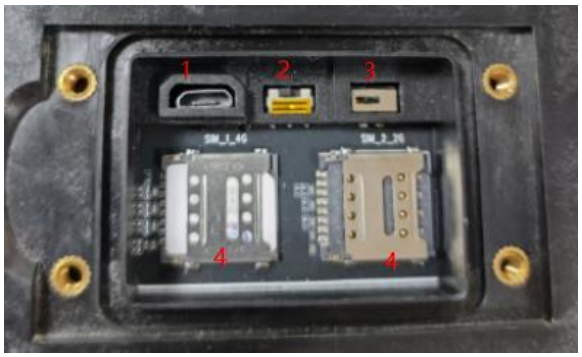
# 5.Device Installation

Before you start install the device, Please check the product you get with the packing list, if you have any problem please contact us or distributor in time.

## 5.1 Prepare works

- ✧ AT10D-MK lock tracker
- ✧ Lock string
- ✧ RFID card
- ✧ 5V/2A DC charger

## 5.2 Interface & switches




① USB for configure;
② Power switch;
③ Casing tamper switch;
④ SIM slot

## 5.3 Lock ON

Just insert the lock wire into the hole, the lock hole will lock on by AUTO.

## 5.4 Unlock(2 Steps)

1. Unlock with RFID/serial/SMS command;
RFID swipe area:

2. Press unlock button to get rope out.

## 5.6 Charger

1. External power charger

OUTPUT: 5V/2A

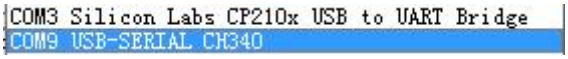
2. Solar panel charging with sunlight

# 6.Configure by serial cable

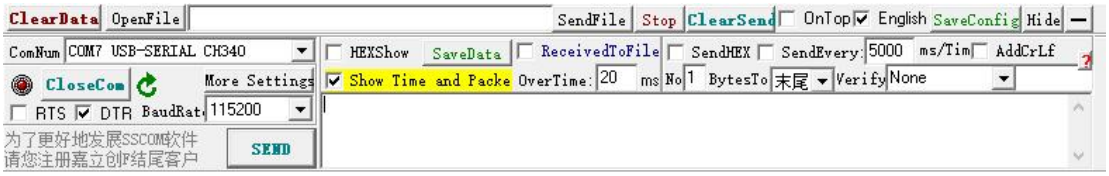
① Set by Tool "SSCOM" or "GPS SSCOM"

② Baud rate: 115200

③ USB Driver:<CH340 USB-SER> or <CP210x_Universal_Windows_Driver>



④ Please use serial command format(Please check at command list for details).




# 7.Setting Commands

To make the device work on GPRS mode together the control center software system, we have to set some parameters to make the device know where and how to connect the server.

| Name | Command | Reply |
|---|---|---|
| SMS SET IP,PORT | AS1234*269#1,120.24.225.253,5577,1# | IP:120.24.225.253,5577,1 OK |
| Set domain | AS1234*269#1,tracking.atelematics.net,5577,1# | |
| SMS SET ID | AS1234*269#2,12007845# | ID: 00012345 OK |
| Set APN | AS1234*269#3,CMNET,,,# | APN:cmnet;OK |
| SMS restart | AS1234*269#B# | Restart;OK |
| SMS set back to factory | AS1234*269#F# | PASSWORD:1234;OK |
| Check IP,ID,APN | AS1234*269#C# | IP:120.24.225.253,2008,1,cmnet,,,ID:61000640,M:6,GPS:A,5,CSQ:28,CGR:1,ACC:1,BAT:33%,VER:AT16-V1.0<br><br>CSQ: GSM signal strength;<br>Ms:6 online<br>(M0:initialize; M2:Search Network; M3:dialing; M4: connecting; M5: login; M6: Online; M9:disconnect and re-connecting) |
| Change angle upload | AS1234*269#G,x# | X: 0-180 degree. |
| SMS query status | AS1234*269#R# | lowBAT:30%,10h ack:0 km:100,5 msg1:24,24 msg2:0,0 |
| SMS set ACC ON/OFF interval | AS1234*269#7,x,y,# | Unit: seconds<br>X=lock on report interval;<br>Y=unlock report interval. |
| SMS set timezone | AS1234*269#I,Ex/Wx,# | X=timezone value<br>E means east timezone;<br>W means west timezone. |

| | | |
|---|---|---|
| | | For example set GMT+5:<br>AS1234*269#I,E5,#<br>If GMT-5:<br>AS1234*269#I,W5,# |
| SMS set Mileage | AS1234*269#m,x# | x=mileage value, Unit: KM |
| Change SMS password | AS1234PASSWORD:4321# | |
| Set authorized number | AS1234*269#5,X,X,X,X,# | X is authorized number,max support 4 numbers. No need country code.<br><br>If the first number is 0, remove it, device will match the phone number compare to the SMS received number.<br><br>For example, number is 0123456789<br>Then cmd is:<br>**AS1234*269#5,123456789#** |
| Query authorized number | AS1234*269#P# | 12345678912,,,,;OK |
| SMS set RFID card white list With group | AS1234*269#6,B,x,ID1,ID2,...ID10,# | Max: 10 ID card support.<br>x=1-10 group, each group max can set 10 RFID numbers,the RFID number need follow with ",";<br><br>Example:<br>Set first 10 ID card<br>AS1234*269#6,B,1,ID1,ID2,ID3,ID4,ID5,ID6,ID7,ID8,ID9,ID10,# |
| Query RFID card number by group | AS1234*269#6,Q,x# | x=1-10 |
| Clear RFID number by group | AS1234*269#6,C,x# | x=1-10 |
| Clear all RFID No. | AS1234*269#6,C# | |
| SMS unlock PW | AS1234*269#8,888888,# | Default unlock PW is 888888 |
| SMS unlock PW | AS1234*269#8,888888,# | Default unlock PW is 888888 |
| Change unlock | AS1234*269#4,888888,123456,# | 6 number PW. |

| | | |
|---|---|---|
| PW | | |
| Check IP,ID,APN | AS1234*269#C# | IP:120.24.225.253,2008,1,cmnet,,,ID:61000640,M:6,GPS:A,5,CSQ:28,CGR:1,ACC:1,BAT:33%,VER:AT16-V1.0 |
| GOOGLE LINK Only support SMS | AS1234WHERE2# | Track from 61000640,PLS help at speed:0Km/h,battery:32%,V, http://maps.google.com/maps?hl=en&q=22.59270,113.87099 2017-08-22 16:04:17<br><br>(M0:initialize; M2:Search Network; M3:dialing; M4: connecting; M5: login; M6: Online; M9:disconnect and re-connecting) |
| OTA command | AS1234*269#W,IP,PORT;# | |
| Sleep mode | AS1234*269#9,M2,x1,x2,# | **M2: close GPS and GSM both, enable sleep with unlock and lock;**<br>x1=wake up working time;<br>x2=sleep mode sleeping time, **need larger than 5min;** |
| Disable sleep mode | AS1234*269#9,M0# | |
| Low power mode | AS1234*269#9,M7,x1,x2,# | M7: Low power enter protection mode(close GSM&GPS), only allow RFID to unlock.<br>X1=Low power percent, default is 10%<br>X2=report duration, unit hour.<br><br>Query low power cmd:<br>***269#R**<br>lowBAT:25%,2h ack:0 vib:100,3,30 km:100,0 msg1:0,0 |
| Exit from deep sleep mode | AS1234*269#9,M7,0# | |
| Clear history data | AS1234*269#2,11223344# | Reset the ID will clear the history data. |
| **LoRa Commands** | | |
| Pair master with slave | AS1234*269#J,B,x,SID1,SID2,...SID10,# | //x=1-2(1 group max set 10 slaves), in total 2 groups max support 14 slaves. |

| | | |
|---|---|---|
| | | For example:<br>AS1234*269#J,B,1,21000010,21000011,#<br><br>If send by SSCOM:<br>*269#J,B,1,21000010,21000011, |
| Query paired slave | AS1234*269#J,Q# | Query all slaves which paired. |
| Query paired slave with group | AS1234*269#J,Q,x# | //x=1-2<br>(1 group max set 10 slaves)<br>1 means group 1;<br>2 means group 2. |
| Clear paired slave with group | AS1234*269#J,C,x# | x=1-2 |
| Clear all slaves | AS1234*269#J,C# | |
| Clear single slave with ID | AS1234*269#J,C,21000010,# | Clear single slave 21000010 |
| Unlock slave via master | AS1234*269#J,A,SID# | For example:<br>AS1234*269#J,A,21000010#<br><br>Can only unlock 1 slave each time. |
| Master Lock lora disconnect duration | AS1234*269#9,M9,x# | X=slave lora disconnect duration, for example x=300, means in 300s, master never received once with slave lora signal then think it's disconnect.<br><br>**SET TO MASTER LOCK.** |
| slave lock lora reporting interval | AS1234*269#9,M9,x,y# | X=lora work duration, y=lora sleep duration, unit seconds.<br>**For example:**AS1234*269#9,M9,10,50#<br>Means slave lora work 10s and sleep 50s by turns. |
| **SMS command format convert to Serial command** | **"AS1234"+"*Serial command*" + "#"**<br><br>**Remove "AS1234" and last "#"**<br>For example, query status SMS command is: AS1234*269#C#<br>If send by serial, then command is: *269#C | |

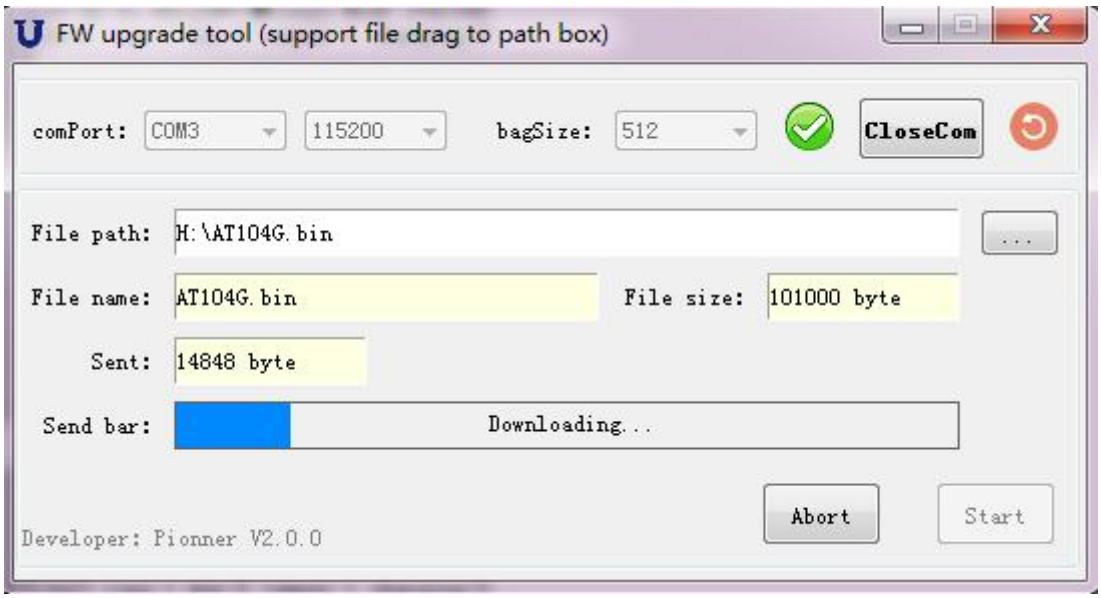
# 8.Upgrade steps

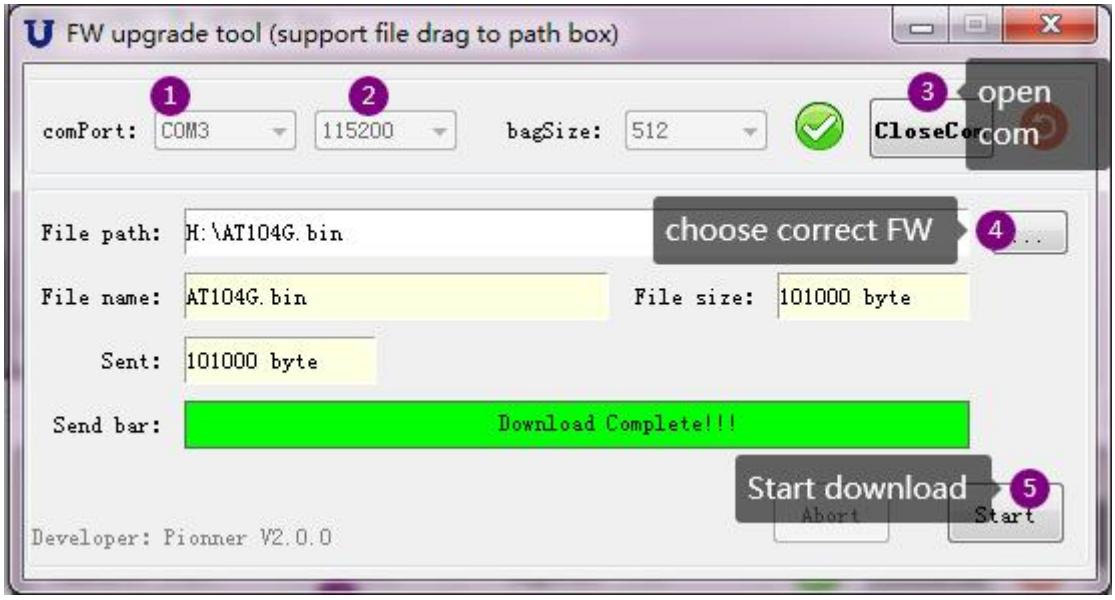
1. Install the eFWTool.Driver:

| | | | |
|---|---|---|---|
| eFWTool Driver | 2012/4/19 17:52 | 应用程序 | 1,721 KB |
| eFWTool | 2022/9/14 14:54 | 应用程序 | 860 KB |

2. serial cable:

3. Power on the device, make sure the device is under working status and battery normal(>30%);
4. Connect device to computer and set the FW upgrade tool as below:






# 9.Bluetooth APP configuration

① Download APP:

② Choose the latest version and install it.

# 10. FAQ

### 8.1 Show offline in Software Platform

- Check if the device LED indicator shows normal. If device is not beside you, use telephone to check. Call the SIM card telephone you inserted in device, and make sure the SIM card is working.
- Check device parameters, make sure IP ID,APN all correct.
- Make sure the SIM card has money/enabled GPRS, and was correct installed, and make sure the device not in a GSM signal blind area (such as underground garage).

### 8.3 Device do not get correct position

First time get position will take longer time (around 1~2 min), please put the LED side face to sky with no cover to get position. If it takes long time and do not get correct location, please check if the device install position is correct and no metal shield above it.